Method for Walk Correction using Leading Edge Slope:

Derived from dual threshold feature of TOFHIR2C

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Discussion draft; not approved for public release.

Abstract

Two discriminator thresholds measure a finite-interval slope on the pulse's leading edge. A linear fit describes the mean slope as a function of charge. Its inverse provides a time-walk model with one amplitude, AWC, per channel.

This study measures slopes in run 4076 and determines AWC by comparison with independent MCP calibration curves from run 4399. The fit uses only charge intervals populated in both runs.

The comparison tests calibration shapes, not corrected-event timing resolution. The present AWC calibration still needs the MCP reference; reference-free determination remains a future step.

Two thresholds measure how fast the pulse rises

Problem: a fixed threshold gives an amplitude-dependent timestamp. A steeper leading edge generally crosses the threshold earlier.

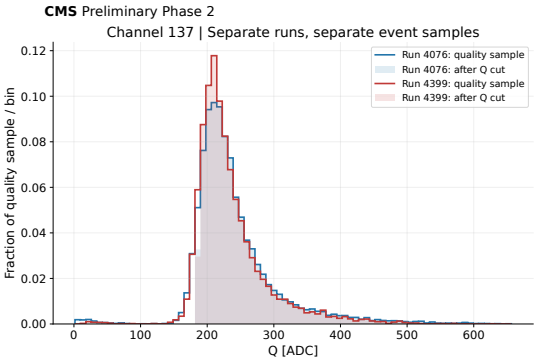
$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

1. Reconstruct both threshold-crossing times with consistent coarse and fine timing.
2. Divide the threshold-current separation by the crossing-time separation.
3. Use charge Q to describe the mean slope for each channel.

S is a finite-interval proxy, not the instantaneous derivative or the current difference itself.

Run 4076 uses $\Delta I = 7.8125\,\mu$A from the implemented threshold conversion. GitHub: timing and threshold reconstruction.

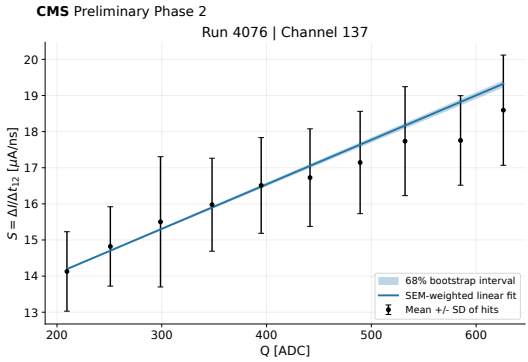
1. Select the prompt population and the charge peak region



1. Require one hit per target-bar end; no other hit in that module.
2. Leave the other module unconstrained; ignore channel 192.
3. Require finite Q, S, positive threshold and time separations, and the same coarse bin.
4. Keep $Q \geq Q_{\mathrm{left}}$.

Channel 137: lines show quality-selected charge spectra; shading adds the charge selection. Q_{left} is the smoothed histogram's left half-height boundary, not a Landau-fit boundary. Run 4076 starts at about 184.9 ADC; no extra upper-charge or 100 ps minimum-rise cut is applied.

2. Fit the mean slope as a function of charge



Group the selected run-4076 hits into ten charge bins.
Fit profile means with SEM weights:

$$\overline{S}(Q) = c_0 + c_1 Q.$$

Use all selected hits, without a training/test split.
Resample hits and refit 1000 times with fixed bin edges.

Channel 137 has 19 126 selected hits. Points show mean slope; bars show the hit standard deviation (SD). SEM is the standard error of the mean. The band is the conditional 68% bootstrap interval of the fitted mean. Linearity remains a working approximation.

3. Convert the fitted slope into a walk correction

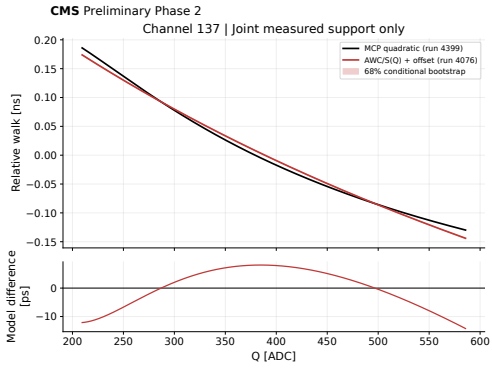
$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

- A is AWC, a current amplitude in μA.
- A is constant for one channel and operating configuration.
- The correction changes event by event through Q.

The slope fit determines the charge dependence, but does not determine AWC. The present calibration obtains AWC from an external MCP reference. Use the model only over its calibrated charge range with a positive slope denominator.

The model concerns the mean pulse slope. Its band is not a single-hit timing uncertainty.

4. Determine AWC where both runs have measured charge support



Run 4076: slope model.
Run 4399: MCP quadratic $W_{\mathrm{MCP}} = a + b/Q + c/Q^2$.
Fit A and a free offset B to match

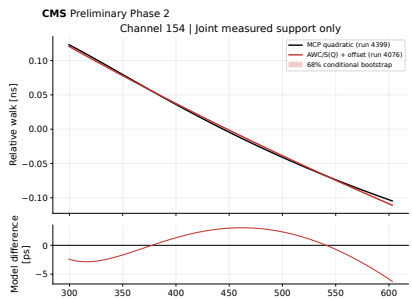
$$\frac{A}{c_0 + c_1 Q} + B$$

to the MCP curve.
Use a uniform charge grid within their populated overlap only.

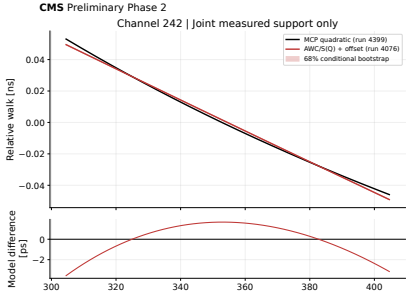
Channel 137: black = MCP reference; red = fitted slope model. The runs contain different events and assume the same nominal setup. AWC = 18.332 μA. The lower panel shows model differences, not event times; RMS mismatch = 6.80 ps.

Apply the same comparison to both detector modules

Upstream: channel 154

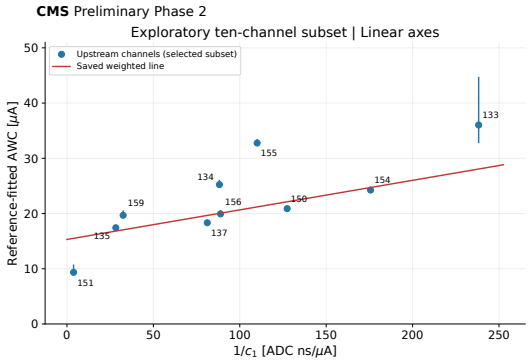


Downstream: channel 242



Each fit uses a separate AWC and offset over the measured overlap. Bands propagate run-4076 slope replicas with the MCP curve fixed. In run 4399, downstream channels require upstream bar 8 in coincidence. Run 4076 leaves the opposite module unconstrained. The opposite module does not supply a timing reference for this fit.

AWC varies between channels beyond a simple slope-coefficient trend



Each point represents one of ten selected upstream channels.
Horizontal coordinate: $1/c_1$.
Vertical coordinate: MCP-calibrated AWC.
The fitted line gives $\chi^2/\mathrm{ndf} = 538.6/8$.
A straight line does not adequately describe this illustrative subset.

Bars are slope-only 16–84% bootstrap intervals. The fit treats x as fixed and uses half the AWC interval as its symmetric error. Reference errors and correlations are omitted. The ten channels were selected for discussion, not by a derived quality rule.

The slope gives the shape; the present AWC still uses MCP

1. Select prompt hits and fit the mean slope versus charge.
2. Form the inverse-slope walk model.
3. Fit AWC and an offset to MCP curves within measured overlap.
4. Apply the charge-dependent correction within the calibrated range.

Uncertainties: displayed bands propagate slope-fit statistics with the MCP curve fixed. They exclude reference-fit, model, selection and cross-run uncertainties.

Next step: validate corrected-event timing and determine AWC from internal information. Shape agreement alone does not demonstrate reference-free calibration.

The full dual-threshold note includes the additional MCP-covariance and charge-weighting checks. GitHub: selection and profiles; GitHub: MCP reference sample.

References

[1] "Performance of the ALICE Time-Of-Flight detector at the LHC", Eur. Phys. J. Plus (2013) 128: 44

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[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[4] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

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[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

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[8] J.E. Grund, SLAC-PUB-2416 (Oct. 1979).

[9] A.R. Frolov, T.V. Oslopova, Yu.N. Pestov, "Double threshold discriminator for timing measurements", Nuclear Instruments and Methods Volume 356, Issues 2/3 (1995) https://doi.org/10.1016/0168-9002(94)01239-3.

[10] This will arise in LGAD that are not thinned to remove the effect (J. Vavra, private communication). See also PICOSEC, "Fast Timing with MicroPattern Gas Detectors", where charge in the electron peak is captured-ignoring the ion tail..

[11] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[12] GitHub repository, https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector.